

GEOFFREY MAINA WACERA
TRANSACTION MONITORING ANALYST | MACHINE LEARNING/AUTOMATION ENGINEER

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Profile

Data Analytics, Automation engineering, and AI-driven professional with 5+ years of experience spanning transaction monitoring, financial data analysis, banking operations, software development, and technology-enabled risk management. Currently responsible for real-time monitoring and investigation of transactional activity across Equity Bank Group subsidiaries, using SQL, PostgreSQL, Python, Grafana, and ETL techniques to identify anomalies, investigate unusual activity, validate monitoring controls and support data-driven risk decisions.

I leverage my background in data, technology and financial operations to identify processes that can be made faster, more reliable and less manual. With experience in SQL, Python, n8n, UiPath, software development and AI, I connect operational problems with practical data and automation solutions, including fraud monitoring, anomaly detection, investigation workflows, analysis and reporting. This combination allows me to understand both how to build a solution and how it can improve day-to-day business operations.

Education

University of Nairobi

Bachelor of Economics
Second Class Honors (Upper Division)

Nairobi, Kenya
December 2021

Nairobi School

KCSE Score: A- (80 points)

Nairobi, Kenya
November 2015

Experience

EQUITY BANK (K) LTD

Transaction Monitoring Analyst

Nairobi, Kenya
August 2024 – Present

- Conduct 24/7 monitoring and real-time analysis of financial transactions across Equity Bank Group subsidiaries in Kenya, Uganda, Tanzania, South Sudan, and Rwanda to identify unusual activity, potential fraud indicators, operational risks, and control weaknesses.
- Analyze transactional trends, patterns, thresholds, and anomalies using SQL, Python, PostgreSQL, and Grafana dashboards to support risk identification, investigation, reporting, and data-driven decision-making.
- Investigate transaction-monitoring alerts using SQL, PostgreSQL, and transactional data analysis to validate alerts, identify unusual patterns, determine root causes, and support financial crime risk decisions.
- Contribute to the development and continuous refinement of the enterprise-wide transaction monitoring system, ensuring accurate and complete transaction capture across core banking, mobile banking, agency, and card channels.
- Evaluate monitoring rules, system logic, thresholds, and alert behavior to improve detection effectiveness, identify control weaknesses, and reduce false-positive alerts.
- Assess data integrity and monitoring-system behavior by validating transaction flows, investigating processing anomalies, and identifying gaps that could affect the effectiveness of financial crime controls.
- Collaborate with business units, compliance teams, technology stakeholders, and Group subsidiaries to investigate high-risk activity and communicate analytical findings, supporting timely and well-evidenced enterprise financial crime decisions.
- Monitor transaction-monitoring system availability and processing performance, proactively coordinating resolution of downtime, processing delays, and monitoring gaps to maintain continuous surveillance.
- Designed automated fraud-analysis workflows that compare transaction activity against hourly, daily, weekly, and monthly behavioural baselines, supporting identification of unusual spikes and emerging transaction patterns.
- Built n8n-based fraud-monitoring and automation workflows integrating SQL, Python, data processing and AI capabilities to automate repetitive investigation and monitoring activities, strengthen alert analysis and improve operational efficiency.
- Mapped repetitive monitoring and investigation activities into automated workflow steps, defining inputs, processing logic, outputs, exception conditions and escalation points to support consistent and efficient execution.
- Developed and executed functional test scenarios for automation workflows, validating data inputs, business logic, workflow execution, exception handling and expected outputs before implementation.
- Partner with business stakeholders to define reporting requirements, improve data transformation processes, and develop dashboards and reports that support risk monitoring, control effectiveness, and decision-making.

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E&M TECHNOLOGY HOUSE LTD

Nairobi, Kenya

Software Developer

January 2024 – July 2024

- Led a cross-functional development team through requirements gathering, process analysis, solution design, development, functional testing, deployment, documentation, and stakeholder validation for an automation solution.
- Developed and integrated backend services and RESTful APIs using Java and Spring Boot, applying software architecture, database integration, Git, and Agile development and integration practices.
- Applied Python, SQL, and database technologies to support data extraction, transformation, validation, automation, and application workflows, with a focus on improving operational efficiency.
- Developed and maintained Android applications using Kotlin, Jetpack Compose, XML, MVVM, Room Database, and LiveData, contributing to application performance, maintainability, and user experience.
- Coordinated sprint planning, task allocation, technical problem-solving, and delivery while mentoring team members and translating business requirements into practical technical solutions.

EQUITY BANK (K) LTD

Nairobi, Kenya

Relationship Officer

May 2021 – December 2023

- Performed customer-facing banking activities in accordance with KYC, AML, CTF, and regulatory requirements, supporting effective customer due diligence and identification of potentially unusual or higher-risk activity.
- Processed domestic and international wire payments in accordance with KYC, AML, CTF, and POCAMLA requirements, ensuring appropriate customer and transaction information was maintained and regulatory controls were followed.
- Achieved a 100% compliance rating on cheque processing and liquidations, maintaining zero violations under applicable Cheque Truncation guidelines.
- Maintained strong customer relationships and reduced account dormancy from 30% to 13% through proactive customer engagement and account management.
- Increased digital-product adoption by 22% through targeted customer engagement and education.
- Delivered an average 95% of daily transaction targets while maintaining a 98% service-level performance based on customer feedback.

Selected Data and AI Project

Finance Analytics Platform | Live Portfolio Project

- Designed, developed, and deployed a finance analytics platform for customer, account, and transaction analysis using Python, SQL, PostgreSQL, Pandas, FastAPI, REST APIs, HTML/CSS/JavaScript and Plotly.
- Built SQL-driven analytics, ETL workflows and an AI assistant using Python, PostgreSQL and AI technologies, demonstrating how data, automation and AI bot development can improve financial analysis and operational efficiency.

Portfolio: finance-analytics-platform-1.onrender.com

Core Skills

Technical Skills:

- **Data and Analytics:** Python, SQL, PostgreSQL, Pandas, Excel, ETL/ELT, data validation, data transformation, exploratory data analysis, anomaly detection, statistical analysis, data visualization
- **AI and RPA:** AWS Certified Machine Learning Engineer – Associate, machine learning workflows, n8n workflows, UiPath, Power Automate, RAG agents, AI-assisted analysis, AI automation
- **Monitoring and Risk Technology:** Grafana, transaction monitoring, fraud detection, AML/CTF, KYC, alert analysis, monitoring rules, thresholds, transaction-data validation, root-cause analysis
- **Software Engineering:** Java, Spring Boot, Kotlin, Jetpack Compose, XML, REST APIs, MVVM, Room Database, Docker, Git, Agile/Scrum

Soft Skills:

- Analytical Thinking and Problem Solving
- Leadership and Team Collaboration
- Attention to Detail and Accuracy
- Communication and Stakeholder Management
- Risk Identification and Assessment
- Financial Crime and Compliance

Languages:

- English (Fluent)
- Swahili (Fluent)
- French (Basic)